

Reciprocal Lerch Convolutions

Entire-order identities, central-factorial zeta reductions,
all logarithmic moments, and Stieltjes–Gamma calculus

Research continuation for Vladimir Reshetnikov’s ProveIt programme
Mathematical derivations and computational checks with OpenAI assistance

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Abstract

A reciprocal-argument product of two Lerch profiles has an ordinary integral equal to $(s+t)\zeta(s+t+1, a)$, interpreted as an entire function of the total order. We develop this order-addition mechanism at every multiplicative depth. An explicit central-factorial polynomial reduces an r -fold convolution to finitely many Lerch values, and at unit fugacity to Hurwitz-zeta or alternating Hurwitz-zeta values. A separate finite differential-polynomial calculus evaluates *every* mixed logarithmic moment for arbitrary complex orders. For two factors we obtain a closed formula whose apparent zeta poles cancel by a polynomial divisibility identity.

The resulting order derivatives give ordinary convergent products representing generalized Stieltjes constants. Higher-depth derivatives involve finite Stirling and generalized-harmonic polynomials, positive odd zeta jets, and an explicit Stieltjes term. Gamma-weighted contour integrals, reciprocal polylogarithmic remainders, a Gaussian odd-part family, unequal-shift partial fractions, and anchored antiderivatives are derived in the same normalization. Negative-order resonances retain nonzero residue contributions; one explicit depth-four term is $1/3$.

The article provides full analytic proofs, exact finite checks, numerical diagnostics, and integration notes. It continues the repository’s Mellin and Stieltjes programme in a distinct reciprocal geometry; it does not claim to settle the same-direction two-scale problem, the Gaussian S_6/S_8 conjectures, or historical priority for the underlying convolution method.

Source anchor. Targeted repository reading is pinned to 0c740647c48baec7f0b4192723e977d69917dd52. Three accessible incoming packages were inspected in relevant parts; three newer binary packages were listed but not inspected. No repository files were modified.

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1 Research target, contribution, and provenance

The canonical ProveIt polylogarithm manuscript contains an extensive theory of harmonic sums, cyclotomic polylogarithms, Hurwitz order derivatives, Stieltjes distributions, and Gamma correlations [1]. Its Mellin–Lerch continuation asks for exact product integrals and their spectral derivatives [2]. The incoming bilinear report treats positive-integer polylogarithm orders under same-direction rational Mellin kernels, with a finite multiple-polylogarithm closure theorem and further questions about arbitrary orders [3].

Here the change in geometry is essential. We study reciprocal arguments and, more generally, multiplicative constraints $y_1 \cdots y_r = e^\mu$. This is not a relabeling of the same-direction integral

$$\int_0^\infty \frac{x^{b-1} \operatorname{Li}_s(-zx) \operatorname{Li}_t(-wx)}{(1+x)^N} dx. \quad (1.1)$$

The reciprocal geometry has a simple bilateral transform, and that transform permits an entire-order finite closure absent from the integer-only cone expansion of the preceding report. Our results solve the explicitly formulated reciprocal problem, not all of (1.1).

1.1 A first identity and the central normalization

Define, for $y > 0$ and $\operatorname{Re} a > 0$,

$$F_{a,s}(y) = y\Phi(-y, s, a), \quad f_{a,s}(x) = F_{a,s}(e^x), \quad \mathcal{E}_a(v) = v\zeta(v+1, a). \quad (1.2)$$

The last function is filled in at $v = 0$ by $\mathcal{E}_a(0) = 1$; it is entire in v . Our normalization gives

$$F_{1,s}(y) = -\operatorname{Li}_s(-y), \quad F_{a,0}(y) = \frac{y}{1+y}. \quad (1.3)$$

The most economical instance of the theory is

$$\boxed{\int_0^\infty F_{a,s}(y) F_{a,t}(1/y) \frac{dy}{y} = \mathcal{E}_a(s+t), \quad s, t \in \mathbb{C}.} \quad (1.4)$$

All the displayed integrals in this article, unless explicitly described as contour integrals, are ordinary absolutely convergent integrals. No distribution product or finite-part cutoff is hidden in (1.4). In particular, the value at $s+t=0$ is 1, not zero and not a directional limit.

The full results go considerably beyond this two-factor identity. Theorems 3.1 and 4.3 treat every number of factors and every complex order. Theorem 5.1 evaluates all mixed powers of their logarithmic arguments by a finite recurrence. Theorem 5.2 gives a particularly short closed two-factor formula. The Stieltjes, harmonic, Gamma, and antiderivative consequences are proved in Sections 6–10.

1.2 Conventions and classical inputs

We use

$$\Phi(z, s, a) = \sum_{n=0}^\infty \frac{z^n}{(n+a)^s}, \quad \zeta(s, a) = \sum_{n=0}^\infty (n+a)^{-s}$$

in their initial domains and their standard analytic continuations. For $\operatorname{Re} a > 0$, $\log(n+a)$ is the branch in the right half-plane. Pochhammer symbols are rising factorials: $(s)_0 = 1$, $(s)_j = s(s+1) \cdots (s+j-1)$. A superscript in square brackets denotes an order derivative, while $\psi^{(k)}$ denotes an argument derivative of the digamma function. Generalized Stieltjes constants have the convention

$$\zeta(1+v, a) = \frac{1}{v} + \sum_{n=0}^\infty \frac{(-1)^n}{n!} \gamma_n(a) v^n; \quad \gamma_0(a) = -\psi(a). \quad (1.5)$$

These conventions agree with the standard Hurwitz, Lerch, Fermi-integral, and Gamma identities summarized in the DLMF [6].

The elementary mechanisms are classical: Euler's beta integral, bilateral convolution, the semigroup law for one-sided fractional integrals, and partial fractions. Real convolution methods for Fermi–Dirac-type integrals already appear explicitly in Selvaggi and Selvaggi [7]. The incoming unequal-twist report also uses resolvent partial fractions, harmonic coefficients, and monodromy, in a periodic Fourier setting [4]. We do not present those methods as discoveries. The contribution here is their explicit entire-order, all-depth, all-logarithmic-moment specialization and the exact identities it produces. We make no claim that the initial two-factor integral, or every special case below, is new in the wider literature.

1.3 What the source audit does and does not establish

The source pin was read through the connected GitHub repository. The canonical README, the Gaussian quotient chapter, and the Mellin research-question section were inspected at that pin. Relevant sections of the incoming bilinear, unequal-twist, and nonlinear-transport packages were also available through the user's Library. Three further incoming packages, named for cubic harmonic mixed orders, exact resonant identities, and independent orders, were listed in the incoming directory but were not available for content inspection. Their complete filenames are recorded in the source-audit artifact. Their names alone establish neither overlap nor mathematical content.

The existing S_4 proof is not counted again. The current short Gaussian S_6 reduction and revised S_8 candidate remain conjectural in the inspected canonical manuscript. Our Gaussian specialization is an ordinary integral family; it is not a zero certificate for either frozen relation vector. No proof-assistant formalization is supplied.

2 Analytic profiles and their bilateral transform

Write $f_0(x) = e^x/(1 + e^x)$. For $\operatorname{Re} s > 0$ the Lerch integral gives

$$f_{a,s}(x) = \frac{1}{\Gamma(s)} \int_0^\infty u^{s-1} e^{-(a-1)u} f_0(x-u) du. \quad (2.1)$$

The right side explains the covariant derivative $D_a = \partial_x + a - 1$:

$$D_a f_{a,s} = f_{a,s-1}. \quad (2.2)$$

The analytic justification of all subsequent parameter operations is concentrated in the next lemma.

Lemma 2.1 (Weighted holomorphy). *Fix $\operatorname{Re} a > 0$ and choose a real c satisfying*

$$\max(0, 1 - \operatorname{Re} a) < c < 1. \quad (2.3)$$

Then $s \mapsto e^{-cx} f_{a,s}(x)$ is an entire family in $L^1(\mathbb{R}) \cap L^\infty(\mathbb{R})$. The same is true after multiplication by any fixed power of x , after any fixed order derivative, and locally jointly in a . The profile is smooth in x , and (2.2) holds for all $s \in \mathbb{C}$.

Proof. Every derivative of f_0 is f_0 times a polynomial in f_0 , so $|f_0^{(j)}(x)| \leq C_j f_0(x)$ on the real line. The function $e^{-cx} f_0(x)$ is bounded and integrable for $0 < c < 1$. For $\operatorname{Re} s > 0$, multiply (2.1) by e^{-cx} and put $v = x - u$. Its absolute integral is bounded by a constant times

$$\int_0^\infty u^{\operatorname{Re} s-1} e^{-(c+\operatorname{Re} a-1)u} du \int_{\mathbb{R}} e^{-cv} f_0(v) dv.$$

Both factors are finite. The same estimate with the second factor a supremum proves boundedness. On compact subsets of the parameter half-planes, derivatives insert only fixed powers of u and $\log u$, so the estimates prove holomorphy in these norms.

For $\operatorname{Re} s > 1$, differentiation and integration by parts in u give $D_a f_{a,s} = f_{a,s-1}$: the derivative of $e^{-(a-1)u} f_0(x-u)$ is its negative covariant x derivative, and both boundary terms vanish. For a compact set of arbitrary complex s , choose an integer N with $\operatorname{Re}(s+N) > 0$ and define $f_{a,s} = D_a^N f_{a,s+N}$. The integration-by-parts identity makes these definitions consistent on overlaps. They agree with the original Lerch series when $x < 0$ and hence with its continuation. The estimates for $f_0^{(j)}$ prove the asserted norm holomorphy everywhere.

Finally choose $\delta > 0$ so that $c - \delta$ and $c + \delta$ both satisfy (2.3). The elementary inequality $|x|^m \leq C_{m,\delta} e^{\delta|x|}$ transfers the two neighboring weighted estimates to every power of x . Cauchy's formula in s and a gives the same conclusions for their derivatives. $\square$

Theorem 2.2 (Bilateral transform). *For $s \in \mathbb{C}$, $\operatorname{Re} a > 0$, and $\max(0, 1 - \operatorname{Re} a) < \operatorname{Re} p < 1$,*

$$\int_{\mathbb{R}} e^{-px} f_{a,s}(x) dx = \Gamma(p)\Gamma(1-p)(p+a-1)^{-s} = \frac{\pi}{\sin \pi p} (p+a-1)^{-s}. \quad (2.4)$$

The power uses the right-half-plane logarithm of $p+a-1$.

Proof. For $\operatorname{Re} s > 0$, Fubini applies to (2.1) by the preceding estimates. The substitution $x = v + u$ separates the transform into

$$\frac{1}{\Gamma(s)} \int_0^\infty u^{s-1} e^{-(p+a-1)u} du \int_{\mathbb{R}} e^{-pv} f_0(v) dv.$$

The first factor is $(p+a-1)^{-s}$. With $y = e^v$, the second is $\int_0^\infty y^{-p}/(1+y) dy = B(1-p, p)$. Euler's reflection formula gives the stated Gamma product. Both sides are entire in s , so the identity theorem completes the proof. $\square$

There is no factor $\Gamma(s)$ left in (2.4). Losing this normalization would destroy order addition and every Stieltjes specialization. The common strip depends on the shift a , not on the spectral order s . This distinction is what makes the all-complex-order statements possible.

3 Order addition at every multiplicative depth

For $r \geq 2$, set $x_r = \mu - x_1 - \cdots - x_{r-1}$ and define

$$(f_{a,s_1} * \cdots * f_{a,s_r})(\mu) = \int_{\mathbb{R}^{r-1}} \prod_{j=1}^r f_{a,s_j}(x_j) dx_1 \cdots dx_{r-1}. \quad (3.1)$$

Equivalently $y_j = e^{x_j}$ and $y_1 \cdots y_r = e^\mu$, with measure $\prod_{j < r} dy_j/y_j$. For $r = 1$ use ordinary evaluation.

Theorem 3.1 (Entire-order addition). *For $\operatorname{Re} a > 0$, real μ , and arbitrary $s_1, \dots, s_r \in \mathbb{C}$, (3.1) is absolutely convergent and depends on the orders only through $W = s_1 + \cdots + s_r$. Denote it by $\mathcal{C}_{r,a}(W; \mu)$. It is entire in W and has the representation*

$$\mathcal{C}_{r,a}(W; \mu) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} e^{p\mu} \left(\frac{\pi}{\sin \pi p} \right)^r (p+a-1)^{-W} dp, \quad (3.2)$$

where c satisfies (2.3). This formula extends it holomorphically to $|\operatorname{Im} \mu| < r\pi$. Every fixed mixed order derivative can be passed through (3.1); a derivative of total degree M gives $\partial_W^M \mathcal{C}_{r,a}(W; \mu)$.

Proof. Put $g_j(x) = e^{-cx} f_{a,s_j}(x)$. The factor $e^{c\mu}$ is constant on the constraint hyperplane. The integral of absolute values in (3.1) is therefore at most

$$e^{c\mu} \|g_r\|_\infty \prod_{j < r} \|g_j\|_1.$$

Lemma 2.1 proves absolute convergence and locally uniform bounds for all parameter derivatives. The Fourier transform of g_j is (2.4) at $p = c + it$. Its product is

$$\left(\frac{\pi}{\sin \pi p} \right)^r (p + a - 1)^{-W}.$$

This depends only on W . Fourier inversion is applicable because the product decays like $e^{-r\pi|t|}$ times a fixed power of $1 + |t|$ on compact order sets. It gives (3.2) and establishes pointwise equality, not merely equality almost everywhere; the convolutions are continuous. The same exponential decay proves the stated μ strip and entire dependence on W . Norm holomorphy and Cauchy's formula justify all differentiated integrals. $\square$

For $\operatorname{Re} W > 0$, another useful form follows from the ordinary one-sided fractional-integral semigroup:

$$\mathcal{C}_{r,a}(W; \mu) = \frac{1}{\Gamma(W)} \int_0^\infty u^{W-1} e^{-(a-1)u} f_0^{*r}(\mu - u) du. \quad (3.3)$$

One can prove it first with all $\operatorname{Re} s_j > 0$ by Fubini and the Dirichlet simplex integral; their sum of positive integration variables has Gamma density of order W . Alternatively its transform is (3.2). The kernel in (3.3) is then continued through $W = 0, -1, \dots$ by the already established entire function, not by substituting into an improper integral that no longer converges.

3.1 A direct two-factor proof

It is useful to verify the main normalization without Fourier inversion. For $\operatorname{Re} s, \operatorname{Re} t > 0$, insert the two Euler integrals and integrate the reciprocal variable first. For $u, v > 0$,

$$\int_0^\infty \frac{dy/y}{(1 + ye^{-u})(1 + y^{-1}e^{-v})} = \frac{u + v}{1 - e^{-(u+v)}}.$$

This follows by one partial fraction decomposition; the logarithms at infinity cancel. With $w = u + v$, the remaining beta integral yields

$$\begin{aligned} \int_0^\infty F_{a,s}(y) F_{a,t}(1/y) \frac{dy}{y} &= \frac{1}{\Gamma(W)} \int_0^\infty \frac{w^W e^{-(a-1)w}}{e^w - 1} dw \\ &= W\zeta(W + 1, a). \end{aligned}$$

Absolute convergence and entire-order continuation proved above give (1.4) for all s, t . In particular,

$$\int_0^\infty \operatorname{Li}_s(-y) \operatorname{Li}_t(-1/y) \frac{dy}{y} = (s + t)\zeta(s + t + 1), \quad (3.4)$$

with the removable value at zero understood. For example, the integrals for the order pairs (1, 1), (2, 1), and (2, 2) are respectively $2\zeta(3)$, $3\zeta(4)$, and $4\zeta(5)$. These examples illustrate the normalization; they are not offered as historical-priority claims.

4 Central-factorial kernels and finite zeta reductions

4.1 The logistic convolution polynomial

Define monic polynomials

$$p_{2m}(X) = X \prod_{j=1}^{m-1} (X^2 + 4j^2\pi^2), \quad m \geq 1, \quad (4.1)$$

$$p_{2m+1}(X) = \prod_{j=1}^m (X^2 + (2j-1)^2\pi^2), \quad m \geq 0. \quad (4.2)$$

Thus $p_1 = 1$, $p_2 = X$, and

$$p_{r+2}(X) = (X^2 + r^2\pi^2)p_r(X). \quad (4.3)$$

These are central-factorial product polynomials after an imaginary rescaling of their variable. Their degree is $r-1$ and their parity is $(-1)^{r-1}$.

Lemma 4.1 (All-depth elementary kernel). *For every $r \geq 1$ and real μ ,*

$$f_0^{*r}(\mu) = \frac{p_r(\mu)}{(r-1)! [1 - (-1)^r e^{-\mu}]}. \quad (4.4)$$

At even depth the quotient at $\mu = 0$ is filled in by continuity.

Proof. At depth one this is f_0 . At depth two the substitution $y = e^x$ gives $f_0 * f_0(\mu) = \mu/(1 - e^{-\mu})$, including its limit 1. Let $g(p) = \pi \csc(\pi p)$. Direct differentiation gives

$$\frac{d^2}{dp^2} g(p)^r = r(r+1)g(p)^{r+2} - r^2\pi^2 g(p)^r.$$

Multiplication by μ^2 corresponds to two derivatives of the bilateral transform. Transform uniqueness therefore implies

$$f_0^{*(r+2)}(\mu) = \frac{\mu^2 + r^2\pi^2}{r(r+1)} f_0^{*r}(\mu).$$

The two initial cases and (4.3) prove the result. All transforms used here lie in the common strip $0 < \operatorname{Re} p < 1$; polynomial multiplication is justified by the neighboring-strip estimates from Lemma 2.1. $\square$

4.2 A finite formula away from the central point

Put $\epsilon_r = (-1)^r$.

Theorem 4.2 (Finite Lerch closure). *For $\mu < 0$, $\operatorname{Re} a > 0$, and $W \in \mathbb{C}$,*

$$C_{r,a}(W; \mu) = \frac{-\epsilon_r e^\mu}{(r-1)!} \sum_{j=0}^{r-1} \frac{(-1)^j p_r^{(j)}(\mu)}{j!} (W)_j \Phi(\epsilon_r e^\mu, W + j, a). \quad (4.5)$$

The combined expression continues to all real μ . At even depth and $\mu > 0$, the continuation of the whole sum is independent of whether the Lerch cut is approached from above or below.

Proof. For $\operatorname{Re} W > 0$, insert (4.4) into (3.3). When $\mu < 0$, expand the denominator as

$$\frac{1}{1 - \epsilon_r e^{u-\mu}} = -\epsilon_r e^{\mu-u} \sum_{n=0}^{\infty} \epsilon_r^n e^{n(\mu-u)}.$$

Expand $p_r(\mu - u)$ by its finite Taylor formula. The remaining integrals are $\Gamma(W + j)(n + a)^{-W-j}/\Gamma(W)$, yielding (4.5). Absolute convergence justifies the steps. Both sides are entire in W for fixed $\mu < 0$.

The left side is holomorphic for $|\operatorname{Im} \mu| < r\pi$. At even depth, the discontinuity of the Lerch integral at $z = e^\mu > 1$ is, up to a common nonzero sign convention, $2\pi i e^{-a\mu} \mu^{W+j-1}/\Gamma(W + j)$. After multiplication by $(W)_j$ and the other factors, the jump of the sum is proportional to

$$\sum_{j=0}^{r-1} \frac{(-1)^j p_r^{(j)}(\mu) \mu^j}{j!} = p_r(0) = 0.$$

This calculation is initially made away from Gamma poles and then continued in W . Thus the two real boundary values agree and coincide with the ordinary convolution. At $\mu = 0$ use the entire convolution, or the central formula below. At odd depth the argument $-e^\mu$ does not cross the positive Lerch cut. $\square$

At depth two and $a = 1$, the formula becomes

$$\int_0^\infty \operatorname{Li}_s(-y) \operatorname{Li}_t(-e^\mu/y) \frac{dy}{y} = W \operatorname{Li}_{W+1}(e^\mu) - \mu \operatorname{Li}_W(e^\mu). \quad (4.6)$$

This is a branch-cancelled combination, not a license to discard the imaginary parts of its terms separately. Its limit at $\mu = 0$ is $\mathcal{E}_1(W)$.

4.3 Pole-free central formulas

Define the entire alternating Hurwitz function

$$\eta(v, a) = \Phi(-1, v, a) = 2^{-v} \left[\zeta\left(v, \frac{a}{2}\right) - \zeta\left(v, \frac{a+1}{2}\right) \right]. \quad (4.7)$$

The two residues at $v = 1$ cancel. In particular $\eta(0, a) = 1/2$ and $\eta(1, a) = \frac{1}{2}[\psi((a+1)/2) - \psi(a/2)]$. Write

$$p_{2m}(X) = \sum_{k=0}^{m-1} A_{m,k} X^{2k+1}, \quad p_{2m+1}(X) = \sum_{k=0}^m B_{m,k} X^{2k}.$$

The coefficients are explicitly the elementary symmetric functions of the squared even or odd multiples of π in (4.1)–(4.2).

Theorem 4.3 (Central finite closure). *For $\operatorname{Re} a > 0$ and every $W \in \mathbb{C}$,*

$$\mathcal{C}_{2m,a}(W; 0) = \frac{1}{(2m-1)!} \sum_{k=0}^{m-1} A_{m,k}(W)_{2k} \mathcal{E}_a(W + 2k), \quad (4.8)$$

$$\mathcal{C}_{2m+1,a}(W; 0) = \frac{1}{(2m)!} \sum_{k=0}^m B_{m,k}(W)_{2k} \eta(W + 2k, a). \quad (4.9)$$

Every summand in these formulas is entire in W .

Proof. For $\operatorname{Re} W > 0$, (3.3) at zero uses $p_{2m}(u)/(e^u - 1)$ in even depth and $p_{2m+1}(u)/(e^u + 1)$ in odd depth. The ordinary Mellin integrals are $(W)_j \zeta(W + j, a)$ or $(W)_j \eta(W + j, a)$. In even depth $j = 2k + 1$, so

$$(W)_{2k+1} \zeta(W + 2k + 1, a) = (W)_{2k} \mathcal{E}_a(W + 2k).$$

This supplies the pole-free form. Entire continuation proves both identities at all remaining orders. $\square$

The first cases, useful for checking every normalization, are

$$\mathcal{C}_{2,a}(W; 0) = \mathcal{E}_a(W), \quad (4.10)$$

$$\mathcal{C}_{3,a}(W; 0) = \frac{1}{2} \{ (W)_2 \eta(W + 2, a) + \pi^2 \eta(W, a) \}, \quad (4.11)$$

$$\mathcal{C}_{4,a}(W; 0) = \frac{1}{6} \{ (W)_2 \mathcal{E}_a(W + 2) + 4\pi^2 \mathcal{E}_a(W) \}, \quad (4.12)$$

$$\mathcal{C}_{5,a}(W; 0) = \frac{1}{24} \{ (W)_4 \eta(W + 4, a) + 10\pi^2 (W)_2 \eta(W + 2, a) + 9\pi^4 \eta(W, a) \}, \quad (4.13)$$

$$\mathcal{C}_{6,a}(W; 0) = \frac{1}{120} \{ (W)_4 \mathcal{E}_a(W + 4) + 20\pi^2 (W)_2 \mathcal{E}_a(W + 2) + 64\pi^4 \mathcal{E}_a(W) \}. \quad (4.14)$$

An equivalent differential-operator form is

$$\mathcal{C}_{r,a}(W; 0) = \frac{1}{(r-1)!} \begin{cases} p_r(-\partial_a) \zeta(W, a), & r \text{ even}, \\ p_r(-\partial_a) \eta(W, a), & r \text{ odd}. \end{cases} \quad (4.15)$$

The even polynomial has no constant term, so it annihilates the a -independent zeta residue.

4.4 Negative-order resonance is not zero substitution

At $W = -2$, (4.12) gives

$$\boxed{\mathcal{C}_{4,a}(-2; 0) = \frac{1}{3} + \frac{2\pi^2}{3} B_2(a), \quad B_2(a) = a^2 - a + \frac{1}{6}.} \quad (4.16)$$

Indeed $(-2)_2 \mathcal{E}_a(0)/6 = 1/3$, while $\mathcal{E}_a(-2) = -2\zeta(-1, a) = B_2(a)$. The uncombined expression $(W)_3 \zeta(W + 3, a)$ has a vanishing Pochhammer factor next to a pole at $W = -2$. Setting the first factor to zero loses the $1/3$. This is a concrete integration safeguard, not an assertion that the inspected repository already printed that error.

5 Every logarithmic moment at arbitrary complex orders

Order addition alone evaluates the unweighted convolution. Polynomial weights record how the total logarithmic argument is distributed among its factors. Surprisingly, they still admit finite closure at arbitrary complex orders.

5.1 An all-depth differential-polynomial algorithm

For $\ell \geq 0$ define polynomials $R_\ell(s; T, U)$ by

$$\begin{aligned} R_0 &= 1, \\ R_{\ell+1} &= (T + sU)R_\ell + (\pi^2 + T^2)\partial_T R_\ell + U^2\partial_U R_\ell. \end{aligned} \quad (5.1)$$

Their first values are

$$\begin{aligned} R_1 &= T + sU, \\ R_2 &= 2T^2 + 2sTU + s(s+1)U^2 + \pi^2, \\ R_3 &= 6T^3 + 6sT^2U + 3s(s+1)TU^2 + (s)_3U^3 + 5\pi^2T + 3s\pi^2U. \end{aligned}$$

For a multi-index $\ell = (\ell_1, \dots, \ell_r)$, let

$$\mathcal{M}_\ell(s; a; \mu) = \int_{\mathbb{R}^{r-1}} \prod_{j=1}^r x_j^{\ell_j} f_{a,s_j}(x_j) dx_1 \cdots dx_{r-1}, \quad x_r = \mu - \sum_{j < r} x_j. \quad (5.2)$$

Theorem 5.1 (Finite closure of all mixed moments). *The integral (5.2) is absolutely convergent and entire in its orders. Form the finite polynomial $\prod_{j=1}^r R_{\ell_j}(s_j; T, U)$ and put $W = \sum s_j$. Apply the following linear replacement rules to its monomials:*

$$T^{2k} U^d \mapsto \sum_{q=0}^k \binom{k}{q} (-\pi^2)^{k-q} \mathcal{C}_{r+2q,a}(W+d; \mu), \quad (5.3)$$

$$T^{2k+1} U^d \mapsto \sum_{q=0}^k \binom{k}{q} \frac{(-\pi^2)^{k-q}}{r+2q} \{ \mu \mathcal{C}_{r+2q,a}(W+d; \mu) - (W+d) \mathcal{C}_{r+2q,a}(W+d+1; \mu) \}. \quad (5.4)$$

The resulting expression is exactly $\mathcal{M}_\ell(s; a; \mu)$. Thus every mixed logarithmic moment at $\mu = 0$ reduces to finitely many entire Hurwitz-zeta combinations in even depth, and alternating Hurwitz-zeta combinations in odd depth. Arbitrary mixed order derivatives of this finite formula are valid.

Proof. Put $g(p) = \pi \csc(\pi p)$, $h = p + a - 1$, $T = \pi \cot(\pi p)$, and $U = h^{-1}$. Then $g' = -Tg$, $T' = -(\pi^2 + T^2)$, and $U' = -U^2$. Repeated differentiation gives the exact identity

$$(-\partial_p)^\ell \{g(p)h^{-s}\} = g(p)h^{-s} R_\ell(s; T, U).$$

The left side is the transform of $x^\ell f_{a,s}(x)$. The product of the transforms of all factors is consequently $g(p)^r h^{-W} \prod_j R_{\ell_j}(s_j; T, U)$.

The identity $T^2 = g^2 - \pi^2$ proves (5.3). For an odd power, first use that identity and then $g^R T = -(g^R)'/R$. Integration by parts on the vertical line in (3.2) gives, for every v ,

$$\frac{1}{2\pi i} \int e^{p\mu} g(p)^R T h^{-v} dp = \frac{\mu}{R} \mathcal{C}_{R,a}(v; \mu) - \frac{v}{R} \mathcal{C}_{R,a}(v+1; \mu).$$

The boundary terms vanish exponentially. This proves the odd rule. Lemma 2.1 supplies absolute convergence and parameter holomorphy, so no formal interchange remains unproved. $\square$

For example, at $\mu = 0$, with $s = s_1$ and $W = \sum s_j$,

$$\mathcal{M}_{(1,0,\dots,0)} = \left(s - \frac{W}{r}\right) \mathcal{C}_{r,a}(W+1; 0), \quad (5.5)$$

$$\begin{aligned} \mathcal{M}_{(2,0,\dots,0)} &= 2\mathcal{C}_{r+2,a}(W; 0) - \pi^2 \mathcal{C}_{r,a}(W; 0) \\ &\quad + \left[s(s+1) - \frac{2s(W+1)}{r}\right] \mathcal{C}_{r,a}(W+2; 0). \end{aligned} \quad (5.6)$$

For two distinct positions having orders s and t ,

$$\begin{aligned} \mathcal{M}_{(1,1,0,\dots,0)} &= \mathcal{C}_{r+2,a}(W; 0) - \pi^2 \mathcal{C}_{r,a}(W; 0) \\ &\quad + \left[st - \frac{(s+t)(W+1)}{r}\right] \mathcal{C}_{r,a}(W+2; 0). \end{aligned} \quad (5.7)$$

Summing (5.5) over all positions gives zero, as it must from $x_1 + \cdots + x_r = 0$. The factor $1/r$ is essential.

5.2 A closed two-factor formula

For $k \geq 1$ define

$$Q_k(s, t) = \frac{(s)_k - (-1)^k(t)_k}{s + t + k - 1}. \quad (5.8)$$

This is a polynomial, not a meromorphic quotient: at $t = 1 - k - s$, $(t)_k = (-1)^k(s)_k$, proving divisibility. Its first values are

$$Q_1 = 1, \quad Q_2 = s - t, \quad Q_3 = s^2 - st + t^2 + s + t.$$

Let d_{2j} be defined by

$$\frac{\pi\lambda}{\sin \pi\lambda} = \sum_{j=0}^{\infty} d_{2j} \lambda^{2j}, \quad d_0 = 1, \quad d_{2j} = 2(1 - 2^{1-2j})\zeta(2j) \quad (j \geq 1). \quad (5.9)$$

Theorem 5.2 (All two-factor logarithmic moments). *For $\ell \geq 0$, $s, t \in \mathbb{C}$, and $\operatorname{Re} a > 0$, put $W = s + t$. Then*

$$\boxed{\int_0^\infty \log^\ell y F_{a,s}(y) F_{a,t}(1/y) \frac{dy}{y} = \ell! \sum_{j=0}^{\lfloor \ell/2 \rfloor} \frac{d_{2j}}{(\ell - 2j + 1)!} Q_{\ell-2j+1}(s, t) \mathcal{E}_a(W + \ell - 2j)}. \quad (5.10)$$

Every term on the right is entire in the two independent orders.

Proof. Introduce the locally holomorphic tilted integral

$$I(\lambda) = \int_0^\infty y^\lambda F_{a,s}(y) F_{a,t}(1/y) \frac{dy}{y}.$$

The weighted estimates prove convergence for $|\operatorname{Re} \lambda| < \min(1, \operatorname{Re} a)$. For $\operatorname{Re} s, \operatorname{Re} t > 0$ and sufficiently small $|\lambda|$, inserting Euler integrals and evaluating the elementary rational y integral gives

$$I(\lambda) = \frac{\pi}{\sin \pi\lambda} \sum_{n=0}^{\infty} \left[(n + a - \lambda)^{-s} (n + a)^{-t} - (n + a)^{-s} (n + a + \lambda)^{-t} \right]. \quad (5.11)$$

For clarity, the elementary integration kernel before the two Laplace integrals are performed is

$$\frac{\pi}{\sin \pi\lambda} \frac{e^{\lambda u} - e^{-\lambda v}}{e^{u+v} - 1} e^{-(a-1)(u+v)}.$$

It is multiplied by $u^{s-1}v^{t-1}/(\Gamma(s)\Gamma(t))$. Expanding the geometric denominator and integrating u, v gives (5.11). Its $\lambda = 0$ value is removable.

Taylor expansion in λ now yields

$$I(\lambda) = \frac{\pi}{\sin \pi\lambda} \sum_{k=1}^{\infty} \frac{(s)_k - (-1)^k(t)_k}{k!} \zeta(W + k, a) \lambda^k.$$

Normal convergence in a small disk follows from $\operatorname{Re} W > 0$ and the shift bound $|\lambda| < \operatorname{Re} a$. Extracting the coefficient of λ^ℓ and using

$$((s)_k - (-1)^k(t)_k) \zeta(W + k, a) = Q_k(s, t) \mathcal{E}_a(W + k - 1)$$

proves (5.10) in the initial domain. Both sides are entire in s, t , which proves the theorem everywhere. $\square$

The first two nonconstant moments are therefore

$$\int_0^\infty \log y F_{a,s}(y) F_{a,t}(1/y) \frac{dy}{y} = \frac{s-t}{2} \mathcal{E}_a(W+1), \quad (5.12)$$

$$\int_0^\infty \log^2 y F_{a,s}(y) F_{a,t}(1/y) \frac{dy}{y} = \frac{s^2 - st + t^2 + s + t}{3} \mathcal{E}_a(W+2) + \frac{\pi^2}{3} \mathcal{E}_a(W). \quad (5.13)$$

In particular, at $s = t = 0$ the second moment is $\pi^2/3$, the ordinary logistic second moment. Formula (5.12) independently checks the $1/2$ in the first-moment normalization. All odd moments vanish when $s = t$, directly reflecting $y \leftrightarrow 1/y$.

6 Ordinary Stieltjes products and harmonic coefficient calculus

6.1 A convergent product at every Stieltjes index

Set

$$L_m^a(y) = \partial_s^m F_{a,s}(y)|_{s=0}. \quad (6.1)$$

These are order derivatives, not argument derivatives. From (1.5),

$$\mathcal{E}_a^{(M)}(0) = \begin{cases} 1, & M = 0, \\ (-1)^{M-1} M \gamma_{M-1}(a), & M \geq 1. \end{cases} \quad (6.2)$$

Theorem 6.1 (Ordinary Stieltjes product law). *For $m, n \geq 0$ and $\operatorname{Re} a > 0$,*

$$\boxed{\int_0^\infty L_m^a(y) L_n^a(1/y) \frac{dy}{y} = \begin{cases} 1, & m = n = 0, \\ (-1)^{m+n-1} (m+n) \gamma_{m+n-1}(a), & m+n > 0. \end{cases}} \quad (6.3)$$

All integrals are absolutely convergent.

Proof. Differentiate (1.4) m times in s and n times in t . Lemma 2.1 justifies the differentiation. The right side is (6.2) with $M = m+n$. $\square$

For example,

$$\int_0^\infty L_1^a(y) L_0^a(1/y) \frac{dy}{y} = -\psi(a), \quad \int_0^\infty L_1^a(y) L_1^a(1/y) \frac{dy}{y} = -2\gamma_1(a).$$

At $a = 1$ these become identities involving only order derivatives of ordinary polylogarithms. The apparent product can have either sign: this is not a Hermitian inner product and the second factor has reciprocal argument rather than complex conjugation.

6.2 Logarithmic Stieltjes identities

Theorem 5.2 can be differentiated independently in its two orders. Three explicit consequences are

$$\int_0^\infty \log y L_1^a(y) L_0^a(1/y) \frac{dy}{y} = \frac{1}{2} \zeta(2, a), \quad (6.4)$$

$$\int_0^\infty \log^2 y L_1^a(y) L_0^a(1/y) \frac{dy}{y} = \frac{\pi^2}{3} \gamma_0(a) + \frac{2}{3} \zeta(3, a), \quad (6.5)$$

$$\int_0^\infty \log^2 y L_1^a(y) L_1^a(1/y) \frac{dy}{y} = \frac{4}{3} \zeta^{[1]}(3, a) - \frac{2\pi^2}{3} \gamma_1(a). \quad (6.6)$$

To check the last identity explicitly, write $Q = s^2 - st + t^2 + s + t$ and differentiate $Q\mathcal{E}_a(s+t+2)$ at $s = t = 0$. The result is $-\mathcal{E}_a(2) + 2\mathcal{E}'_a(2) = 4\zeta^{[1]}(3, a)$. The remaining term is $\pi^2 \mathcal{E}_a''(0)/3$. This derivation fixes both the coefficient $4/3$ and the Stieltjes sign.

6.3 All-depth jets as finite harmonic polynomials

For a multi-index $\mathbf{m}$ of length r , let $J_{r,\mathbf{m}}(a)$ be the integral at $\mu = 0$ of the product of $f_{a,s_j}^{[m_j]}(x_j)$ evaluated at all $s_j = 0$. Let $M = \sum m_j$. Order addition proves that this integral depends only on M , so write it as $J_{r,M}(a)$. The next result makes that value completely explicit.

Theorem 6.2 (All-depth Stieltjes coefficients). *At even depth, for $M \geq 1$,*

$$J_{2m,M}(a) = \frac{1}{(2m-1)!} \left[A_{m,0}(-1)^{M-1} M \gamma_{M-1}(a) + \sum_{k=1}^{m-1} A_{m,k} \sum_{h=1}^{\min(M,2k+1)} \frac{M!}{(M-h)!} \begin{bmatrix} 2k+1 \\ h \end{bmatrix} \zeta^{[M-h]}(2k+1, a) \right]. \quad (6.7)$$

Here $\begin{bmatrix} j \\ h \end{bmatrix}$ is an unsigned Stirling number of the first kind. For $M = 0$, $J_{2m,0} = A_{m,0}/(2m-1)!$. At odd depth the corresponding formula is

$$J_{2m+1,M}(a) = \frac{1}{(2m)!} \sum_{k=0}^m B_{m,k} \sum_{h=0}^{\min(M,2k)} \frac{M!}{(M-h)!} \begin{bmatrix} 2k \\ h \end{bmatrix} \eta^{[M-h]}(2k, a), \quad (6.8)$$

where $\begin{bmatrix} 0 \\ 0 \end{bmatrix} = 1$. In particular $J_{2m+1,0} = B_{m,0}/[2(2m)!]$.

Proof. Differentiate (4.8) and (4.9) M times at zero. In even depth keep the $k = 0$ term in the entire form $A_{m,0}\mathcal{E}_a(W)$. For $k \geq 1$ it is safe to use $(W)_{2k+1}\zeta(W+2k+1, a)$. The identity $(W)_j = \sum_h \begin{bmatrix} j \\ h \end{bmatrix} W^h$ and the ordinary product rule give (6.7). The same expansion gives (6.8). The zero-order formulas follow immediately. $\square$

For example,

$$J_{4,1}(a) = \frac{2\pi^2}{3} \gamma_0(a) + \frac{1}{3} \zeta(3, a), \quad (6.9)$$

$$J_{4,2}(a) = -\frac{4\pi^2}{3} \gamma_1(a) + \zeta(3, a) + \frac{2}{3} \zeta^{[1]}(3, a). \quad (6.10)$$

The number of factors and the distribution of the derivative orders can be varied without changing these scalar values, provided their respective totals stay fixed.

The coefficients also have a finite generalized-harmonic description. For $j \geq 1$,

$$(z)_j = (j-1)!z \exp \left(\sum_{q=1}^{\infty} \frac{(-1)^{q-1}}{q} H_{j-1}^{(q)} z^q \right), \quad (6.11)$$

locally at zero. Thus, for $1 \leq h \leq j$,

$$\partial_z^h (z)_j \Big|_0 = h(j-1)! \mathcal{B}_{h-1} \left(H_{j-1}, -H_{j-1}^{(2)}, 2!H_{j-1}^{(3)}, \dots \right), \quad (6.12)$$

where $\mathcal{B}_n$ is the complete exponential Bell polynomial. Indeed, divide $(z)_j$ by $(j-1)!z$, take the logarithm of $\prod_{q=1}^{j-1} (1+z/q)$, and expand. This proves the formula, including the signs and factorials. At nonzero resonances one should use the finite polynomial $(z)_j$ itself, rather than a logarithmic derivative divided by a vanishing factor.

Together with Theorem 5.1, these formulas give a terminating algorithm for every mixed order jet of every logarithmic moment. The coefficient operations are finite; they do not presuppose arithmetic independence of the resulting zeta or Stieltjes values.

7 Reciprocal remainders and Gaussian odd parts

7.1 An all-order identity for polylogarithmic remainders

For $N \geq 0$ define the finite Taylor remainder

$$\mathcal{R}_{N,s}(y) = \text{Li}_s(-y) - \sum_{k=1}^N \frac{(-y)^k}{k^s}. \quad (7.1)$$

The empty sum at $N = 0$ is zero. The elementary Lerch shift formula implies

$$F_{N+1,s}(y) = -(-y)^{-N} \mathcal{R}_{N,s}(y). \quad (7.2)$$

Although each remainder can grow polynomially on the positive real axis, the reciprocal product is integrable.

Corollary 7.1 (Harmonic-tail product identity). *For every $N \geq 0$ and $s, t \in \mathbb{C}$,*

$$\boxed{\int_0^\infty \mathcal{R}_{N,s}(y) \mathcal{R}_{N,t}(1/y) \frac{dy}{y} = \mathcal{E}_{N+1}(s+t) = (s+t) \left[\zeta(s+t+1) - H_N^{(s+t+1)} \right]}. \quad (7.3)$$

The last expression is filled in by its removable value 1 when $s+t=0$. Every order derivative and every logarithmic moment is obtained by substituting $a = N+1$ into the preceding theorems.

Proof. The two powers $(-y)^{-N}$ and $(-1/y)^{-N}$ in (7.2) multiply to one. Apply (1.4), followed by the Hurwitz shift identity $\zeta(v, N+1) = \zeta(v) - H_N^{(v)}$. Normal convergence already proved for the Lerch profiles gives all differentiated versions. $\square$

At $N = 1$ and $s = t = 1$, this gives the elementary-looking but nontrivial identity

$$\int_0^\infty [\log(1+y) - y][\log(1+1/y) - 1/y] \frac{dy}{y} = 2\zeta(3) - 2. \quad (7.4)$$

The factors must remain combined. Expanding their product and integrating the resulting pieces independently introduces divergent integrals and is not a valid proof of the formula.

The Stieltjes constants at the shifted integer arguments also reduce by a finite harmonic-logarithmic correction:

$$\gamma_n(N+1) = \gamma_n(1) - \sum_{k=1}^N \frac{\log^n k}{k}. \quad (7.5)$$

It follows by comparing Laurent coefficients of $\zeta(1+v, N+1) = \zeta(1+v) - \sum_{k=1}^N k^{-1-v}$. For $n = 0$ it reads $\gamma_0(N+1) = \gamma - H_N$. Thus the entire remainder family contains finite generalized harmonic numbers and their order derivatives without changing the convergence normalization.

7.2 A fourth-root polylogarithm family

For complex s and positive y , define the analytic odd part

$$A_s(y) = \frac{\text{Li}_s(iy) - \text{Li}_s(-iy)}{2i}. \quad (7.6)$$

For real s this agrees with an imaginary part. For complex s , (7.6), rather than $\text{Im Li}_s(iy)$, is required for holomorphic order differentiation. The initial series gives

$$A_s(y) = \sum_{n=0}^{\infty} \frac{(-1)^n y^{2n+1}}{(2n+1)^s}, \quad F_{1/2,s}(y^2) = 2^s y A_s(y),$$

with continuation to the positive real axis.

Corollary 7.2 (Gaussian reciprocal order addition). *For all $s, t \in \mathbb{C}$, with $W = s + t$,*

$$\boxed{\int_0^\infty A_s(y) A_t(1/y) \frac{dy}{y} = 2^{-W-1} \mathcal{E}_{1/2}(W) = (1 - 2^{-W-1}) \mathcal{E}_1(W).} \quad (7.7)$$

Every mixed order derivative is an ordinary convergent integral.

Proof. Use the displayed relation to $F_{1/2,s}$ and set $z = y^2$. The measure contributes $1/2$ and the order prefactors contribute 2^{-W} . Theorem 3.1 at depth two gives the first equality. The second follows from $\zeta(v, 1/2) = (2^v - 1)\zeta(v)$, with the pole cancelled before evaluation. $\square$

Since $A_1(y) = \arctan y$, a basic specialization is

$$\int_0^\infty \arctan y \arctan(1/y) \frac{dy}{y} = \frac{7}{4} \zeta(3). \quad (7.8)$$

More characteristic of the order-derivative continuation is the identity

$$\int_0^\infty A_0^{[1]}(y) A_0^{[1]}(1/y) \frac{dy}{y} = -\gamma_1 + \gamma \log 2 - \frac{1}{2} \log^2 2. \quad (7.9)$$

Here $A_0^{[1]} = \partial_s A_s|_{s=0}$. It follows by taking the second derivative at zero of $(1 - 2^{-W-1}) \mathcal{E}_1(W)$.

These identities use fourth-root arguments, but their relation to the repository's Gaussian depth-two periods is not automatic. In particular, (7.7) is not a reduction of $\sum_{n \geq 0} (-1)^n H_n / (2n+1)^6$ in the prescribed $g_{a,b}$ basket. That distinction is retained in the integration status.

8 Gamma-weighted integrals and explicit polygamma bridges

When $\operatorname{Re} a > 1/2$, the symmetric vertical line $p = 1/2 + it$ in (3.2) is admissible. Put $b = a - 1/2$. Euler reflection then gives

$$\boxed{\begin{aligned} \mathcal{C}_{r,a}(W; 0) &= \frac{1}{2\pi} \int_{-\infty}^\infty \left[\Gamma\left(\frac{1}{2} + it\right) \Gamma\left(\frac{1}{2} - it\right) \right]^r (b + it)^{-W} dt \\ &= \frac{1}{2\pi} \int_{\mathbb{R}} \left(\frac{\pi}{\cosh \pi t} \right)^r (b + it)^{-W} dt. \end{aligned}} \quad (8.1)$$

The logarithm of $b + it$ is the right-half-plane branch. For $0 < \operatorname{Re} a \leq 1/2$ use a nonsymmetric line satisfying (2.3); simply keeping the symmetric line can cross $\operatorname{Re}(p + a - 1) = 0$ and is not the stated formula.

Theorem 4.3 now evaluates every integral in (8.1) by a finite Hurwitz-zeta expression. For $r = 2$ and $M \geq 1$, differentiation in W at zero gives

$$\boxed{\frac{\pi}{2} \int_{\mathbb{R}} \frac{\log^M(a - \frac{1}{2} + it)}{\cosh^2 \pi t} dt = -M \gamma_{M-1}(a).} \quad (8.2)$$

There is no alternating sign on the right: the $(-1)^M$ from the power derivative cancels all but a minus sign in (6.2). This is a consistency bridge to classical Stieltjes integral representations, not a claim of first discovery of such representations.

8.1 An ordinary log-Gamma identity

At $W = -1$, differentiation of $\mathcal{E}_a(W) = W\zeta(W + 1, a)$ yields

$$\mathcal{E}'_a(-1) = \zeta(0, a) - \zeta^{[1]}(0, a).$$

Using (8.1) and $\zeta^{[1]}(0, a) = \log \Gamma(a) - \frac{1}{2} \log(2\pi)$ gives

$$\frac{\pi}{2} \int_{\mathbb{R}} \frac{(b + it) \log(b + it) - (b + it)}{\cosh^2 \pi t} dt = \log \Gamma(a) - \frac{1}{2} \log(2\pi). \quad (8.3)$$

All differentiations are dominated by the hyperbolic exponential weight. The classical Hurwitz–Gamma identity is the only additional input [6].

8.2 Higher-depth Gamma and polygamma identities

The all-depth jets in Section 6 give, for example,

$$\frac{1}{2\pi} \int_{\mathbb{R}} \left(\frac{\pi}{\cosh \pi t} \right)^4 \log(b + it) dt = \frac{2\pi^2}{3} \psi(a) + \frac{1}{6} \psi^{(2)}(a), \quad (8.4)$$

$$\frac{1}{2\pi} \int_{\mathbb{R}} \left(\frac{\pi}{\cosh \pi t} \right)^4 \log^2(b + it) dt = -\frac{4\pi^2}{3} \gamma_1(a) + \zeta(3, a) + \frac{2}{3} \zeta^{[1]}(3, a). \quad (8.5)$$

The first uses $\psi^{(2)}(a) = -2\zeta(3, a)$. Positive integer total orders likewise reduce to polygamma values by

$$\zeta(k, a) = \frac{(-1)^k}{(k-1)!} \psi^{(k-1)}(a), \quad k \geq 2.$$

At integer shifts these combine with $\zeta(k, N+1) = \zeta(k) - H_N^{(k)}$; at rational shifts they give finite root-of-unity coordinates after the usual residue-class filter.

One can also evaluate every polynomial t moment in (8.1). Expand $it = (b + it) - b$ to obtain

$$\frac{1}{2\pi} \int_{\mathbb{R}} \left(\frac{\pi}{\cosh \pi t} \right)^r (it)^q (b + it)^{-W} dt = \sum_{j=0}^q \binom{q}{j} (-b)^{q-j} \mathcal{C}_{r,a}(W - j; 0). \quad (8.6)$$

Subsequent order derivatives insert arbitrary powers of $\log(b + it)$. Thus the same finite central-factorial family controls Gamma weights, polynomial moments, polygamma values, and Stieltjes order derivatives.

9 Unequal shifts: finite reductions and arithmetic collapses

Different shifts replace the single resolvent in (3.2) by a product. For $\operatorname{Re} a_j > 0$ define

$$\mathcal{K}(\mathbf{s}, \mathbf{a}; \mu) = (f_{a_1, s_1} * \cdots * f_{a_r, s_r})(\mu).$$

A common c with $\max_j(0, 1 - \operatorname{Re} a_j) < c < 1$ exists. The weighted proof gives

$$\mathcal{K} = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} e^{p\mu} g(p)^r \prod_{j=1}^r (p + a_j - 1)^{-s_j} dp, \quad g(p) = \pi \csc(\pi p). \quad (9.1)$$

This is an entire function of the independent orders.

9.1 A precise rigidity statement

Proposition 9.1 (Rigidity of pure order addition). *Group equal shifts into distinct values a_ν , with grouped orders S_ν . Put $W = \sum_\nu S_\nu$. For a proposed shift b with $\text{Re } b > 0$, the identity*

$$\mathcal{K}(\mathbf{s}, \mathbf{a}; \mu) = \mathcal{C}_{r,b}(W; \mu)$$

holds on an open interval of real μ if and only if every grouped order at a shift different from b is zero and the grouped order at b is W ; absent groups are interpreted as zero. In particular, for strictly positive real orders it forces all shifts to equal b .

Proof. Analyticity in the μ strip turns interval equality into equality on the whole real axis. Taking the bilateral transform, cancelling g^r , and logarithmically differentiating gives the rational identity

$$\sum_\nu \frac{S_\nu}{p + a_\nu - 1} = \frac{W}{p + b - 1}.$$

Equality of residues at distinct poles gives the necessity. Conversely, the stated grouped cancellations give equality of the multipliers with the chosen branches and hence equality of convolutions. Equivalently, a possible multiplicative constant after logarithmic differentiation is one by the behavior as real $p \rightarrow +\infty$ of the resolvent product. $\square$

This is a functional rigidity statement within a specified family. It is not an arithmetic-independence theorem or a prohibition on other special-value reductions. Broader finite-jet nonclosure by monodromy has already been studied for the periodic unequal-twist family [4]; that preceding result is not claimed anew here.

9.2 Dirichlet mixtures and integer-order partial fractions

For $\text{Re } s_j > 0$, the classical Dirichlet parameter formula gives

$$\mathcal{K}(\mathbf{s}, \mathbf{a}; \mu) = \frac{\Gamma(W)}{\prod_j \Gamma(s_j)} \int_{\sum_{j=1}^r u_j \geq 0} \prod_j u_j^{s_j-1} \mathcal{C}_{r, \sum u_j a_j}(W; \mu) du_1 \cdots du_{r-1}. \quad (9.2)$$

Indeed apply the Gamma integral to each resolvent in (9.1), separate the total integration variable from its simplex proportions, and then invert the transform. The real parts of the convexly combined shifts remain positive, so Fubini is justified by a common weighted bound.

If the grouped orders are positive integers m_ν , the mixture has a finite alternative. For distinct $a_1, \dots, a_d$ set

$$c_{\nu,k} = \frac{1}{(m_\nu - k)!} \frac{d^{m_\nu-k}}{dz^{m_\nu-k}} \prod_{\rho \neq \nu} (z + a_\rho - a_\nu)^{-m_\rho} \Big|_{z=0}. \quad (9.3)$$

The elementary partial fraction identity and transform uniqueness give

$$\boxed{\mathcal{K}(\mathbf{m}, \mathbf{a}; \mu) = \sum_{\nu=1}^d \sum_{k=1}^{m_\nu} c_{\nu,k} \mathcal{C}_{r,a_\nu}(k; \mu).} \quad (9.4)$$

The number r is the number of original profile factors, not the number of distinct shifts after grouping. At zero, the right side is finite polygamma or alternating-polygamma data. Colliding shifts are handled by the limit of the entire combined expression or by grouping first; individual partial fraction coefficients need not have finite limits.

For instance,

$$\int_0^\infty F_{a,1}(y) F_{b,1}(1/y) \frac{dy}{y} = \frac{\zeta(2, a) - \zeta(2, b)}{b - a}. \quad (9.5)$$

At $b = a$ the value is $2\zeta(3, a)$. For unequal nonnegative integers N, M , this specializes to the finite harmonic quotient

$$\int_0^\infty F_{N+1,1}(y) F_{M+1,1}(1/y) \frac{dy}{y} = \frac{H_M^{(2)} - H_N^{(2)}}{M - N}. \quad (9.6)$$

In particular the values for $(N, M) = (0, 1)$ and $(0, 2)$ are 1 and $5/8$, respectively.

9.3 All-depth elementary integrals with finite harmonic values

Unit-order profiles at integer shifts are elementary logarithmic remainders by (7.2). Their multivariable convolutions can have especially simple arithmetic values. For distinct nonnegative integers $N_1, \dots, N_r$, put

$$c_\nu = \prod_{\rho \neq \nu} (N_\rho - N_\nu)^{-1}, \quad \overline{H}_N^{(q)} = \sum_{j=1}^N \frac{(-1)^{j-1}}{j^q},$$

and let $\mathcal{K}_N$ denote $(f_{N_1+1,1} * \dots * f_{N_r+1,1})(0)$.

Theorem 9.2 (Integral-shift arithmetic collapse). *For $r = 2m \geq 2$,*

$$\mathcal{K}_N = -\frac{1}{(2m-1)!} \sum_{k=0}^{m-1} A_{m,k} (2k+1)! \sum_{\nu=1}^{2m} c_\nu H_{N_\nu}^{(2k+2)}. \quad (9.7)$$

It belongs to $\mathbb{Q}[\pi^2]$, of degree at most $m-1$ in π^2 . For $r = 2m+1 \geq 3$, if all N_ν have the same parity and $\sigma = (-1)^{N_1}$, then

$$\mathcal{K}_N = -\frac{\sigma}{(2m)!} \sum_{k=0}^m B_{m,k} (2k)! \sum_{\nu=1}^{2m+1} c_\nu \overline{H}_{N_\nu}^{(2k+1)}. \quad (9.8)$$

This belongs to $\mathbb{Q}[\pi^2]$, of degree at most m .

Proof. The unit-order partial fraction formula is $\mathcal{K}_N = \sum c_\nu \mathcal{C}_{r, N_\nu+1}(1; 0)$. Moreover $\sum c_\nu = 0$ for $r \geq 2$, by comparing coefficients of p^{-1} at infinity in the resolvent identity.

At even depth insert the central zeta formula and $\zeta(q, N+1) = \zeta(q) - H_N^{(q)}$. The common zeta terms cancel because $\sum c_\nu = 0$, giving (9.7). At odd depth use

$$\eta(q, N+1) = (-1)^N \{ \eta(q, 1) - \overline{H}_N^{(q)} \}.$$

The common parity makes the sign independent of ν , so the common alternating-zeta terms cancel as well. This proves (9.8). All remaining harmonic numbers are rational, and the degrees in π^2 follow from the product polynomials. $\square$

Two explicit instances are

$$(f_{1,1} * f_{3,1} * f_{5,1})(0) = \frac{1475}{13824} + \frac{5\pi^2}{192}, \quad (9.9)$$

$$(f_{1,1} * f_{2,1} * f_{3,1} * f_{4,1})(0) = \frac{575}{3888} + \frac{11\pi^2}{162}. \quad (9.10)$$

These are, respectively, ordinary two- and three-dimensional integrals of explicitly elementary logarithmic profiles. Their derivation requires no integer-relation search.

9.4 A closed consecutive-shift harmonic formula

Define complete homogeneous harmonic polynomials $h_j^{(n)}$ by

$$\prod_{q=1}^n (1 - z/q)^{-1} = \sum_{j=0}^{\infty} h_j^{(n)} z^j, \quad h_j^{(n)} = \frac{1}{j!} \mathcal{B}_j(H_n, 1!H_n^{(2)}, 2!H_n^{(3)}, \dots). \quad (9.11)$$

These have positive harmonic signs, in contrast to the elementary product in (6.11).

Corollary 9.3 (Consecutive even-depth evaluation). *For $r = 2m$ and $n = 2m - 1$,*

$$(f_{1,1} * f_{2,1} * \dots * f_{2m,1})(0) = \frac{1}{n(n!)^2} \sum_{k=0}^{m-1} A_{m,k} (2k+1)! h_{2k+1}^{(n)}. \quad (9.12)$$

Proof. In (9.7), the shifts are $N_\nu = 0, 1, \dots, n$ and $c_N = (-1)^N / [N!(n-N)!]$. Interchanging two finite sums gives

$$\sum_{N=0}^n c_N H_N^{(q)} = \frac{1}{n!} \sum_{j=1}^n \frac{(-1)^j}{j^q} \binom{n-1}{j-1} = -\frac{h_{q-1}^{(n)}}{n n!}.$$

For the last step use $\binom{n-1}{j-1} = j \binom{n}{j} / n$ and

$$\sum_{j=1}^n (-1)^{j-1} \binom{n}{j} j^{-v} = h_v^{(n)}.$$

The latter identity follows by the partial fraction decomposition of $\prod_{q=1}^n (1 - z/q)^{-1}$ and comparison of its z^v coefficient. Substitution proves (9.12). The Bell expression in (9.11) follows by taking a logarithm of its finite product. $\square$

Thus the reciprocal convolution calculus naturally contains both harmonic coefficient systems: the elementary product governing order derivatives of rising factorials, and the reciprocal product governing these finite harmonic collapses. Neither system needs to be inferred from a numerical pattern.

10 Anchored antiderivatives and a finite derivative–primitive algebra

10.1 The covariant fugacity ladder

The transform immediately relates changing total order to integrating or differentiating the logarithmic product argument.

Theorem 10.1 (Covariant primitives and shift derivatives). *For $\operatorname{Re} a > 0$, $W \in \mathbb{C}$, and real μ ,*

$$(\partial_\mu + a - 1) \mathcal{C}_{r,a}(W; \mu) = \mathcal{C}_{r,a}(W - 1; \mu), \quad (10.1)$$

$$\mathcal{C}_{r,a}(W + 1; \mu) = e^{-(a-1)\mu} \int_{-\infty}^{\mu} e^{(a-1)v} \mathcal{C}_{r,a}(W; v) dv, \quad (10.2)$$

$$\partial_a^n \mathcal{C}_{r,a}(W; \mu) = (-1)^n (W)_n \mathcal{C}_{r,a}(W + n; \mu), \quad n \geq 0. \quad (10.3)$$

The primitive in (10.2) is fixed by the condition $e^{(a-1)\mu} \mathcal{C}_{r,a}(W + 1; \mu) \rightarrow 0$ as $\mu \rightarrow -\infty$. All identities can be differentiated in W .

Proof. In (3.2), $\partial_\mu + a - 1$ multiplies the integrand by $p + a - 1$, proving (10.1). Differentiating the power $(p + a - 1)^{-W}$ in a gives (10.3). The contour bound for real negative μ is $O(e^{c\mu})$ on compact order sets. Since $c + \operatorname{Re} a - 1 > 0$, multiplication by $e^{(a-1)\mu}$ tends to zero at $-\infty$ and the integral in (10.2) converges. Integrating (10.1) at order $W + 1$ proves the anchored formula. The same bounds with logarithmic factors justify all order derivatives. $\square$

At $a = 1$, this is an ordinary primitive ladder in μ , or an Euler primitive ladder in $q = e^\mu$ with measure dq/q . Raising W by any positive integer supplies repeated primitives with all constants fixed. At other shifts the integrating factor is indispensable.

Let $D_a = \partial_\mu + a - 1$ and let J_a denote the anchored integral operator on the right of (10.2). On the linear span of the convolution profiles and their order derivatives,

$$[\partial_a, D_a] = 1, \quad [\partial_a, J_a] = -J_a^2. \quad (10.4)$$

The first identity is immediate. For the second, differentiation of the integrating factor gives

$$(\partial_a J_a - J_a \partial_a)h(\mu) = - \int_{-\infty}^{\mu} (\mu - v) e^{-(a-1)(\mu-v)} h(v) dv = -J_a^2 h(\mu).$$

Absolute convergence justifies the repeated integral. Thus a derivative in the shift does not commute with a covariant primitive; the correction is an explicitly normalized second primitive.

Combining (10.3) with order differentiation gives

$$\partial_a^n \partial_W^M \mathcal{C}_{r,a}(W; \mu) = (-1)^n \sum_{h=0}^{\min(M,n)} \binom{M}{h} \partial_W^h (W)_n \partial_W^{M-h} \mathcal{C}_{r,a}(W + n; \mu). \quad (10.5)$$

The coefficients at $W = 0$ are precisely the finite harmonic polynomials in (6.12). Equations (10.4)–(10.5), together with the reciprocal harmonic product in (9.11), provide a concrete compatible derivative–primitive algebra in this family. This addresses a specified instance of the compatibility question in the Mellin continuation [2], not an assertion that every harmonic calculus in the repository has thereby been unified.

10.2 Anchored integration in the Hurwitz shift

For $\operatorname{Re} a, \operatorname{Re} b > 0$, integration can be taken along any path in the right half-plane. A pole-free fundamental identity is

$$\int_a^b \mathcal{E}_c(W) dc = \zeta(W, a) - \zeta(W, b). \quad (10.6)$$

It follows from $\partial_c \zeta(W, c) = -W \zeta(W + 1, c)$. The right side is entire in W , since the residues at $W = 1$ cancel. The central even-depth analogue is

$$\int_a^b \mathcal{C}_{2m,c}(W; 0) dc = \frac{1}{(2m-1)!} \sum_{k=0}^{m-1} A_{m,k}(W) {}_{2k}[\zeta(W + 2k, a) - \zeta(W + 2k, b)]. \quad (10.7)$$

Every bracket is taken as an entire combined difference before specialization.

At odd depth, put $\Delta\eta(v) = \eta(v, a) - \eta(v, b)$. Then

$$\begin{aligned} \int_a^b \mathcal{C}_{2m+1,c}(W; 0) dc &= \frac{1}{(2m)!} \left[B_{m,0} \frac{\Delta\eta(W-1)}{W-1} \right. \\ &\quad \left. + \sum_{k=1}^m B_{m,k}(W) {}_{2k-1} \Delta\eta(W + 2k - 1) \right]. \end{aligned} \quad (10.8)$$

The first quotient is removable at $W = 1$, because $\eta(0, a) = \eta(0, b)$. Its value there is $\eta^{[1]}(0, a) - \eta^{[1]}(0, b)$. These two formulas follow by differentiating their right sides and using the anchor $b = a$; they therefore specify the integration constants as well as the derivatives.

10.3 Stieltjes antiderivatives and Gamma normalization

Taking order derivatives of (10.6) at zero gives

$$\int_a^b \gamma_0(c) \, dc = \log \Gamma(a) - \log \Gamma(b), \quad (10.9)$$

and, for every $n \geq 0$,

$$\int_a^b \gamma_n(c) \, dc = \frac{(-1)^n}{n+1} [\zeta^{[n+1]}(0, a) - \zeta^{[n+1]}(0, b)]. \quad (10.10)$$

Equivalently an indefinite primitive is $(-1)^{n+1} \zeta^{[n+1]}(0, a)/(n+1)$ plus a constant. These are the classical Hurwitz order-derivative primitives, now attached to the ordinary product identities of Theorem 6.1. We do not count the familiar $n = 0$ log-Gamma formula as a new result.

The odd-depth removable primitive also has a finite Gamma value:

$$\eta^{[1]}(0, a) = \log \Gamma(a/2) - \log \Gamma((a+1)/2) - \frac{1}{2} \log 2.$$

This follows from (4.7) and the classical value $\zeta^{[1]}(0, a)$. In particular, the only potentially singular coefficient in (10.8) is resolved explicitly by Gamma differences rather than an unspecified integration constant.

11 Audit findings, verification, and integration status

11.1 Mathematical status and source overlap

No new false equation was identified in the sampled canonical claims or the incoming results used as context. Accordingly this delivery does not manufacture an erratum. It contributes the proved reciprocal identities above and records the exact normalization errors that would arise from certain invalid simplifications.

The incoming bilinear article's same-direction integer-order master [3] and the present all-complex-order reciprocal master are complementary. The former's arbitrary-order same-direction and minimal-depth questions are not solved merely by replacing its kernel with ours. The unequal-twist article [4] already establishes substantial periodic mixed-jet results; our elementary rigidity and partial-fraction arguments should be understood as their ordinary real-line counterparts, not as a new general monodromy theorem. The nonlinear-transport article [5] concerns coordinate-dependent finite parts, whereas no finite-part transport is used in the present convergent convolution family.

Claim	Status in this delivery
Reciprocal all-depth, all-complex-order closure	Proved, with weighted holomorphy and finite formulas.
Every mixed logarithmic moment	Proved by a terminating polynomial recurrence; two-factor closed form supplied.
Stieltjes products, harmonic lattice collapses, and anchored primitives	Proved exact identities.
Pure order-addition rigidity for unequal shifts	Proved within the specified functional family.
Canonical Gaussian S_4	Already proved in the source; unchanged.
Canonical Gaussian S_6 and revised S_8	Not proved or disproved here.
Historical priority for every displayed identity	Not claimed.
Proof-assistant or interval certification	Not supplied.

11.2 Concrete safeguards and corrected intermediate work

The following distinctions are necessary for a correct integration.

Retain entire resonance products. Use $\mathcal{E}_a(v)$ and the even-depth form (4.8), rather than evaluating a vanishing Pochhammer symbol separately from a zeta pole. Formula (4.16) loses $1/3$ under that invalid operation.

Retain the first-moment factor. The correct first moment is $(s - t)\mathcal{E}_a(s + t + 1)/2$. An intermediate calculation during this work omitted the $1/2$. The independent transform argument and the tilted-series coefficient both detect the error. It has been corrected in the final formulas and is retained as a deliberately failing negative control in the code. This was an error in intermediate work, not an error attributed to the repository.

Do not differentiate an integer-only theorem in its integer index. The unequal-shift partial fraction formula holds for specified positive integer orders. Its finite number of terms does not itself extend holomorphically when those integers are replaced by independent complex orders. Use (9.1) or the appropriate Dirichlet mixture for that operation. In contrast, the common-shift moment formula (5.10) is genuinely entire in its independent orders.

Keep branch-cancelled sums combined. At even depth and positive μ , the individual Lerch terms in (4.5) lie on a cut. Their jump cancels because $p_r(0) = 0$. A separated branch assignment that ignores that calculation is not justified. At $0 < \operatorname{Re} a \leq 1/2$, use an admissible nonsymmetric vertical line rather than the Gamma-weighted line $p = 1/2 + it$.

Keep remainders and derivative types distinct. The product in (7.3) is integrable even when its separately expanded pieces are not. Also, $\partial_s \operatorname{Li}_s$, $\partial_y \operatorname{Li}_s$, and $\operatorname{Im} \operatorname{Li}_s$ are different operations. The analytic Gaussian odd part (7.6) is required at complex order.

11.3 Executed computational checks

The delivered script `code/verify.py` executed 336 exact symbolic assertions, 91 floating-point diagnostics at 55 working decimal digits, and 3 negative controls. The exact checks use rational polynomial algebra, with $P = \pi^2$ treated as a formal variable. The numerical diagnostics use independent contour and real-integral evaluations, not fitted integer relations.

Exact assertion family	Count
Central products, parity, and cosecant differential equation	36
Two-factor pole-cancellation and reciprocity polynomials	32
Weighted-transform differential polynomials	9
Stirling versus Bell-harmonic derivatives	136
Unequal-shift partial fraction examples	4
Negative-order residue term	1
First-moment conservation	11
Independent all-depth versus two-factor moment reductions	25
Consecutive-shift complete-harmonic coefficients	80
Explicit triple and quadruple lattice values	2
Total exact assertions	336

The numerical families include complex and negative orders at depths one through eight; shifts below $1/2$ on nonsymmetric contours; direct convolutions of elementary logistic kernels; direct reciprocal polylogarithm moments, including a noninteger-order example; nonzero fugacity on both sides of the apparent positive cut; Stieltjes and log-Gamma integrals; the two unequal-shift lattice examples; and the arctangent and first harmonic-remainder integrals.

Evidentiary limit. These are floating-point diagnostics, not rigorous enclosures. Working precision is not an asserted error bound. The harmonic-remainder quadrature uses a finite logarithmic cutoff, which is recorded in the script. The analytic proofs, not the small residuals, establish the infinite and continuum assertions. The complete machine-readable results, thresholds, package versions, and generated coefficient tables are included in `data/`.

11.4 Integration recommendation

The package is additive. A natural home beneath the Mellin report is

`continuations/reciprocal-lerch-convolutions/`

or a dedicated report path chosen by the maintainer. All theorem and equation labels have the prefix `rlc::`. The integration note contains a short proposed manuscript insertion, a theorem index, the source pin, and explicit conjecture-status wording. Neither the repository nor the user's persistent Library was changed.

12 Further research questions and topics

The following questions seek exact identities or structural classifications. They are not claims inferred from small numerical samples. The first four are especially direct continuations of the proved formulas.

- Q1. A full multivariate tilt formula.** Theorem 5.1 evaluates every Taylor coefficient of $\int e^{\sum_j \lambda_j x_j} \prod_j f_{a,s_j}(x_j)$ under $\sum x_j = \mu$. Find a compact analytic generating function in all λ_j , with its collision limits explicit. The two-factor series (5.11) and the moment-conservation identity supply nonnegotiable normalization checks.
- Q2. Exact unequal-shift mixed order jets.** Use (9.1) to construct a normally convergent expansion into one central shift, including every independent order derivative. The periodic analogue already exists in the incoming unequal-twist report; the real-line version should state its own admissible contour, convergence domain, and continuation across shift collisions. Identify which special combinations still have finite closure without confusing that question with integer partial fractions.
- Q3. Odd integral-shift cancellation sets.** For odd r , the common alternating-zeta coefficient in the unit-order lattice calculation is $S(\mathbf{N}) = \sum_{\nu} c_{\nu}(-1)^{N_{\nu}}$. Classify the distinct integer shift sets for which this rational number is zero. Equal parity is a proved sufficient condition. A classification would produce additional elementary multivariate integrals in $\mathbb{Q}[\pi^2]$. Conversely, $S \neq 0$ alone would not prove arithmetic independence of the surviving odd-zeta combination.
- Q4. A central-factorial normal form for moment identities.** Determine the exact linear relations, as functions of W, a, μ , among the finite families $\mathcal{C}_{r+2q,a}(W + d; \mu)$ produced by the moment algorithm. The product recurrence and covariant ladder already create relations. A canonical finite reduction would improve symbolic output without invoking conjectural independence of special values.

- Q5. Commensurate dilations of logarithmic arguments.** Replace profiles by $f_{a_j, s_j}(q_j x)$ for positive rational q_j . Their transforms have cosecant factors with different periods. Investigate whether a common-denominator decomposition and the Gamma multiplication formula yield finite Hurwitz families at integer orders, and determine which all-complex-order sectors retain a finite closure. The equal-dilation result here is the required base case.
- Q6. Complex shift boundaries and explicit contour contacts.** Continue a beyond the right half-plane while tracking the resolvent branch point and the poles of the cosecant factor. Derive exact residue and branch-integral corrections when a chosen contour is crossed. The ordinary formula for $\text{Re } a > 0$ must remain separate from any new finite-part convention at the boundary.
- Q7. Higher anchored primitives at unequal collisions.** Construct simultaneous primitives in several independent shifts, with all constants fixed on a chosen base point and with confluent limits stated before partial-fraction simplification. The commutator $[\partial_a, J_a] = -J_a^2$ gives a fully proved one-shift model; the multishift problem should identify the corresponding mixed correction operators rather than declare them to commute.
- Q8. The same-direction Mellin problem beyond integer orders.** Return to (1.1). Determine a spectral representation that specializes to the incoming finite double-Hurwitz master but is holomorphic in both orders. The reciprocal geometry provides a useful solvable comparison, not a proof that the same finite basis works in the same-direction geometry.
- Q9. The missing Gaussian functional relation.** Seek a convergent higher-depth or octahedral relation proving the canonical S_6 vector and then the revised S_8 vector. The formal single-product quotient in the manuscript already supplies a separating functional, so enlarging only that same row span cannot work. A useful connection to the present Gaussian integral family would have to produce the missing mixed-color relation explicitly.
- Q10. Noninteger convolution depth.** The contour expression $g(p)^\rho(p+a-1)^{-W}$ can be studied for noninteger ρ after a branch is fixed. Determine the appropriate real kernel and exact special-function expression, and identify precisely why the finite central-factorial product is special to integer depth. No finite-closure impossibility should be asserted without specifying the proposed target function class.
- Q11. Proof-assistant decomposition and literature reconciliation.** Formalize the polynomial recurrences, divisibility of Q_k , harmonic coefficient identities, and lattice partial fractions independently of the analysis. Then isolate weighted Banach-space holomorphy and Fourier inversion as analytic obligations. Before assigning priority labels, compare the exact statement set with the three incoming packages whose contents were not inspected and with the classical convolution literature.

Conclusion

The reciprocal constraint exposes a total-order multiplier whose cosecant and resolvent factors yield central-factorial kernels and Hurwitz order addition. Their interaction gives a finite calculus at every integer depth and every complex spectral order, including all logarithmic moments. Stieltjes products, harmonic remainders, Gamma weights, arithmetic shift collapses, and anchored primitives follow with their normalization and resonance terms intact.

References

- [1] V. Reshetnikov, *ProveIt: polylogarithm manuscript*, canonical manuscript, repository snapshot 0c740647c48baec7f0b4192723e977d69917dd52. Inspected files include Analysis/Polylogarithms/docs/manuscript/README.md and chapters/04-cyclotomic-quotients.tex. <https://github.com/VladimirReshetnikov/ProveIt>.
- [2] ProveIt research continuation, the *Mellin–Lerch dilation continuation*, especially sections/06-audit-research.tex, at the same source pin. Repository path: Analysis/Polylogarithms/docs/reports/stieltjes-correlation-calculus/continuations/mellin-lerch-dilation/.
- [3] Research continuation prepared for ProveIt, *Bilinear Mellin–Hurwitz Calculus: Double-zeta reductions, harmonic moments, and Stieltjes reciprocity*, 10 October 2026. Incoming package ProveIt_Bilinear_Mellin_Hurwitz_2026-10-10.zip; relevant source sections 07–09 inspected from the Library copy.
- [4] Research continuation prepared for ProveIt, *Unequal-Twist Stieltjes Jets: Monodromy Obstructions, Convergent Harmonic Expansions, Gamma Convolutions, and Multishift Zeta Finite Parts*, 10 October 2026. Incoming package ProveIt_Unequal_Twist_Jets_2026-10-10.zip; relevant source statements and scope inspected from the Library copy.
- [5] Research continuation prepared for ProveIt, *Nonlinear Stieltjes Transport: Finite-part cocycles, harmonic contact terms, Möbius averages, and exact polylogarithmic identities*, 10 October 2026. Incoming package ProveIt_Nonlinear_Stieltjes_Transport_2026-10-10.zip; abstract and source scope inspected.
- [6] NIST, *Digital Library of Mathematical Functions*, §§5.5, 5.12, 25.2, 25.11, 25.12, and 25.14, accessed 10 October 2026. In particular: Euler reflection and beta integrals; Hurwitz shift and argument-derivative identities; $\zeta'(0, a)$; the Lerch integral; and the Mellin–Barnes representation 25.14.8. <https://dlmf.nist.gov/>.
- [7] J. P. Selvaggi and J. A. Selvaggi, “The Application of Real Convolution for Analytically Evaluating Fermi–Dirac-Type and Bose–Einstein-Type Integrals,” *Journal of Complex Analysis* **2018** (2018), Article ID 5941485, 8 pages. <https://doi.org/10.1155/2018/5941485>. The convolution and Laplace-inversion methodology is an antecedent; no claim of exhaustive overlap analysis is made.